Day -15

प्रश्नों की संख्या दीजिए

Candidate must not write on margin

Q1 a.) What was the average price of soap in the year 2013 (₹/unit)?

Ans Total sales of soap by x in 2013 = ₹ 3,50,000
Total sales of soap by y in 2013 = ₹ 4,50,000
Total sales of soap by z in 2013 = ₹ 4,00,000

Total unit of x = 3,50,000 / 7 = $\boxed{50,000}$
Total unit of y = 4,50,000 / 5 = $\boxed{90,000}$
Total unit of z = 4,00,000 / 8 = $\boxed{50,000}$

Required average = $\dfrac{\text{Total sales in 2013}}{\text{Total unit sold in 2013}}$

$= \dfrac{3,50,000 + 4,50,000 + 4,00,000}{50,000 + 50,000 + 90,000}$

$= \dfrac{1,2,00,\cancel{000}}{1,90,\cancel{000}} = ₹\boxed{6.315}$ /unit

b.) What was the percentage decrease in unit sold for x in 2012 over 2011?

Ans: Total unit sold for x in 2011 = $\dfrac{\text{Total sales (2011)}}{\text{unit price}}$

$= 3,00,000 / 5 = \boxed{60,000}$ units

Total unit sold for x in 2012 = $\dfrac{\text{Total sales (2012)}}{\text{unit price}}$

$= \dfrac{3,00,000}{6}$

$= \boxed{50,000}$ units.

apple
apple
Publication

प्रश्नों की संख्या दीजिए

Candidate must not write on margin

Required percentage decrease $= \frac{(60,000 - 50,000)}{60,000} \times 100$

$= \frac{10,000}{60,000} \times 100$

$= \boxed{16.66\ \%}$

c) What was the percentage share of Z in the total units sold in 2013?

Ans. Units sold by X in 2013 $= \frac{\text{Total sales of X}}{\text{unit Price}}$

$= \frac{350000}{7}$

$= \boxed{50,000}$ units.

Units sold by Y in 2013 $= \frac{\text{Total sales of Y (2013)}}{\text{unit Price}}$

$= \frac{4,50,000}{5}$

$= \boxed{90,000}$ units

Units sold by Z in 2013 $= \frac{\text{Total sales of Z (2013)}}{\text{unit price}}$

$= \frac{4,00,000}{8}$

$= \boxed{50,000}$ units

apple
Publication

प्रश्नों की संख्या दीजिए

Candidate must not write on margin

Total units sold of soap in 2013 = 50,000 + 50,000 + 90,000

= $\boxed{1,90,000}$ units

Required % share of z in the total unit sold = $\frac{50,000}{1,90,000} \times 100$

= $\boxed{26 \cdot 31\%}$

d) In which year Y sold the maximum number of units?

Ans Units solds by Y in 2011 = $\frac{\text{Total sales (2011)}}{\text{unit price}}$

= $\frac{4\,40,000}{4}$

= $\boxed{1,10,000}$ units

Units sold by Y in 2012 = $\frac{\text{Total sales (2012)}}{\text{unit price}}$

= 420000 / 6

= $\boxed{70,000}$ units

Unit sold by Y in 2013 = $\frac{\text{Total sales (2013)}}{\text{unit price}}$

= 4 50,000 / 5

= $\boxed{90,000}$ units

Unit sold by Y in 2014 = $\frac{\text{Total sales (2014)}}{\text{unit price}}$

$= 490000/7$

$= \boxed{70,000}$ units

~~Total~~ Unit sold by Y in 2015 $= \dfrac{\text{Total sales (2015)}}{\text{unit price}}$

$= \dfrac{420000}{6}$

$= \boxed{70,000}$ units

Maximum number of units sold by Y is in the year $\boxed{2011}$ i.e of $\boxed{1,10,000}$ units.

e.) The maximum price per unit given in the table exceeds the minimum price per unit by what percentage?

Maximum price per unit is by company X in 2015 i.e $\boxed{₹9/\text{unit}}$

Minimum price per unit is by company Y in 2014 i.e $\boxed{₹4/\text{unit}}$

Required % $= \dfrac{9-4}{4} \times 100 = \dfrac{5}{4} \times 100 = \cancel{80\%}$

$= \cancel{1250\%}$ $\boxed{125\ \%}$